FBO1 – CHAMBER FERMION LOCALITY AND LOCAL INTEGRATION

Abstract. This focused working outline develops the chamber estimates needed for local fermion integration. The light and heavy modes, analytic inverse, weighted Gaussian identity and local determinant are stated with their measures and field domains. A single linked overview connects the proof tasks to the uniform local-integration theorem.

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1. Introduction and target statement

FBO1 is the second multiscale proof project within the RGO1 stage of QEO1. Its conclusions supply the fermion inputs to normalization and the nonlinear renormalization step. The first statement gives the conditional endpoint; later pages specify each estimate and its source.

Broader outlines. RGO1: renormalized construction and non-Gaussianity; QEO1: the full construction.

FT. Uniform chamber locality and exact weighted block integration

Model-specific. Overview Target.

Setting. Fix $0 < \rho < 1$, an odd $L \geq 5$, $0 < \alpha < 1$, $\varepsilon > 0$, and a power M of L with $M \geq L^2$. The chamber stencil, normalized pairings, covariant averages, and field domains are defined in F0; its site has four spin components and 48 chamber labels. Work at mesh $\eta = L^{-k}$, integer $k \geq 0$, on a torus with $\ell/(LM)$ odd and at least 13, with fermions antiperiodic in each coordinate, and with $0 < \mu \leq 1$. These are independent level data.

Let a', γ, C_V be the fixed zero-field constants supplied by S02, with $\gamma L \leq 1/6$, and put $q(p) = 54\pi C_V a' p/M$, $\alpha_0 = 1/54$. For fixed $\kappa, \kappa_0 > 0$ take $\kappa_{\mathrm{w}} = \kappa + \kappa_0 + 3$ and $\kappa' = \kappa_{\mathrm{w}} + \kappa_0$. Assume

$$M > 4L, \quad q(M^{27\alpha_0}) \leq \tfrac{1}{2}, \quad q(e^{27\kappa'}) \leq \tfrac{1}{2}, \quad 0 < e_k \leq e_0, \quad 6Le_0^{\varepsilon} \leq 1, \quad 4C_V a' e_0^{\varepsilon} \leq 1.$$

Here e_k is the charge occurring in the hopping phase $e^{ie_k \eta A(c,c')}$; μ is the positive reference mass. For the determinant difference assertion below require additionally, with $v_k = -\log e_k$,

$$v_k \geq 9, \quad e_k^{\varepsilon} \leq \tfrac{1}{4}, \quad 8M(v_k/4 + 4)e_k^{\varepsilon} \leq 1, \quad \ell \geq M v_k^2, \quad e_k^{1/4} + e_k^{\varepsilon} < 1.$$

Claim. For every connection in the local real-gauge domain $\widetilde{\mathcal{G}}_k$ of F0, the penalized fine inverses $S_y = F_y(A)^{-1}$, defined in S02, exist and are analytic in A. The auxiliary parameter is $y \in \mathbb{R} \cup \{\infty\}$ for $k \geq 1$, and $y \in \mathbb{R}$ for $k = 0$. Their local walk and polymer decompositions satisfy

$$N_\gamma(S_y) \leq 32a' w_k(y), \qquad N_\gamma(\delta_{\alpha,A} S_y) \leq 304a', \qquad w_k(y) = \begin{cases} 1 & k \geq 1, \\ (1+|y|)^{-1} & k = 0. \end{cases}$$

The weakened families, for $|s_\square| \leq M^{\alpha_0}$, preserve active polymer components and depend only on their data. Their propagator terms obey

$$\|\mathbf{1}_{\Delta_z} S_y(X,A) \mathbf{1}_{\Delta_{z'}}\| \leq 32a' e^{27\kappa'} w_k(y) e^{-\gamma d(z,z') - \kappa' d_M(X)}.$$

At $k = 0$ the same bound holds for the normalized chamber kernel; short covariant quotients at positive levels include distinct labels in one chamber cluster.

The coarse covariance Γ_k, field maps $H_k, \bar{H}_k$, and local Dirac terms $M_k(X)$ have the exact identities, analytic locality, and uniform bounds of S06. For every polynomial of the Grassmann fields, Gaussian block integration is exactly the weighted completion formula of S07. Its determinant factor is $\delta Z_k = \det(D_k + b_L P_L)$ at positive levels and $\delta Z_0 = \det(D_A + \mu + b_L J P_L J^*)$ at level zero. The incoming integration has four weight-one components per unit site for $k \geq 1$ and 192 components of weight $1/48$ per photon site for $k = 0$; the identity Gaussian field weight is respectively 1 and $\sqrt{48}$.

There are local analytic determinant terms with

$$\frac{\delta Z_k(A)}{\delta Z_k(0)} = \exp \sum_X (E_k^d(X,A) - E_k^d(X,0)), \qquad |E_k^d(X,A)| \leq C_d M^3 e^{-(\kappa+1) d_M(X)}.$$

They are gauge invariant, even, and satisfy analytic reality. For $\mathscr{A} \in \widetilde{R}_k$ and $\mathscr{C} \in \mathfrak{X}_k$, the explicitly defined domains in F0,

$$|E_k^d(X, \mathscr{A} + \mathscr{C}) - E_k^d(X, \mathscr{C})| \leq 8 C_d M^3 e_k^{1/4} e^{-\kappa d_M(X)}.$$

All constants are independent of level, mesh, volume, permitted connection and $\mu \in (0,1]$; they may depend on the fixed ρ, L, α, M and decay parameters. The separate locality estimates in S01 and S02 hold for arbitrary unitary period characters. The transpose and even determinant assertions here use the stated antiperiodic convention.

Proof work. The internal chamber gap separates four light components from the remaining components. Its Schur complement gives a free singular kernel whose mesh-scaled regularity permits local inversion. Fold local cubes, construct both parametrices, and sum their walks. Attach weakening parameters to the full support of every walk. Then form coarse resolvents by the two exact fine resolvent identities. Weighted Berezin completion identifies the integration and its determinant; inserting the local walks in a resolvent integral localizes that determinant. The small-field gain follows from polymer decay on large sets and a local real gauge followed by Cauchy's formula on the remaining sets.

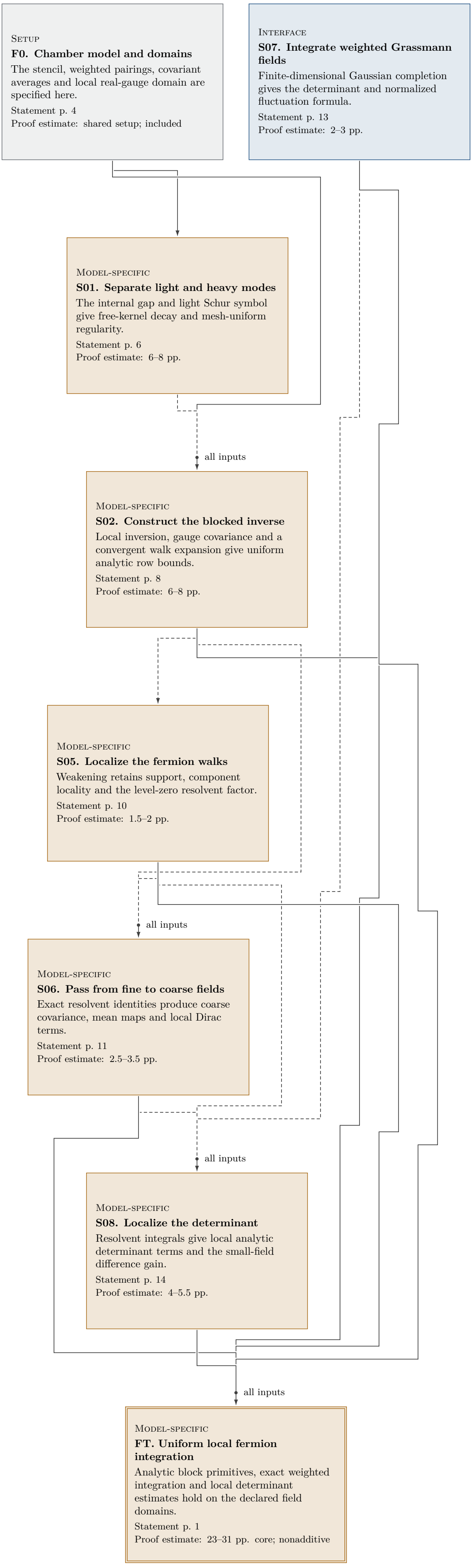
Inputs and scope. The proof uses the chamber model, the geometric facts in F0, and finite Fourier and exterior algebra. In an RG application, the assertion that a photon reconstruction plus its allowed fluctuation belongs to these field domains is an imported Maxwell result, stated explicitly in F0; it is not a consequence of the fermion bounds. This theorem does not tune a mass or construct a nonlinear RG step.

Source. Frozen QEM1, Sections 9 and 11, pp. 77–89 and 102–113; TeX 6779–7808 and 9072–10116. The six supporting cards preserve QEO1 identifiers S01, S02, S05 (fermion clauses), S06, S07, S08. The 23–31-page working estimate covers this conditional fermion project, counting common setup once.

FBO1 – Chamber fermion locality and local integration

Definitions/setup | Model-specific | Interface (general)

Focused outline under RGO1, within QEO1. The model and field domains are in F0. S07 is an independent finite-algebra theorem; its application uses the chamber estimates.



Solid arrows supply inputs to an application; dashed arrows identify use inside a proof. Every incoming branch at a junction is required together with the standing hypotheses on the statement page. Click a box for its statement. Return links are navigation, not proof dependencies.

To evaluate these results on photon reconstructions and fluctuations, the Maxwell domain estimates must also be verified. They are recalled in F0 and proved in MBO1. The six numbered source statements retain their QEO1 identifiers; S05 here contains only its fermion clauses. Page estimates overlap through shared definitions and imports.

2. F0. Chamber spinors, connection domains, and imported photon bounds

Setup. Overview Target.

Chamber stencil and measures. For the local-integration theorem, fermions are antiperiodic in each coordinate of the torus. The broader character range in S01 and S02 applies to their separate estimates. A site is $x = (c, g)$, where $c \in \eta\mathbb{Z}^3$ and g is one of the 48 signed permutation matrices. The spin space is $\mathbb{C}^4$, with Hermitian matrices $\gamma_j = \sigma_3 \otimes \sigma_j$ and $\gamma_5 = \sigma_1 \otimes 1_2$; the Pauli matrices are given in S01. Set

$$x^{(i)} = (c, gs_i)\ (i = 0, 1, 2), \qquad x^{(3)} = (c + \eta g e_1, g r_{e_1}), \quad n = (e_3, (e_2 - e_3)/\sqrt{2}, (e_1 - e_2)/\sqrt{2}, -e_1).$$

Here s_0 changes the third sign, s_1 interchanges coordinates 2 and 3, s_2 interchanges 1 and 2, and r_{e_1} changes the first sign. For $w = (6, 6\sqrt{2}, 6\sqrt{2}, 6)$ put

$$M_i(c, g) = \tfrac{1}{2} w_i(\rho + \gamma \cdot g n_i), \qquad (D_A f)(x) = -\eta^{-1} \sum_{i=0}^{3} M_i(x)[U_A(x, x^{(i)}) f(x^{(i)}) - f(x)].$$

Internal transports are one; the external transport is $e^{ie_k \eta A(c, c+\eta g e_1)}$. The chamber measure is $\eta^3/48$ and the photon-site spinor measure is η^3. Thus $J\xi(c, g) = \xi(c)$ is isometric and $J^* f = 48^{-1} \sum_g f(c, g)$. The chamber identity kernel is $48\eta^{-3}\mathbf{1}[x = x']1_4$.

Unit and L-blocks are centered odd cubes. The covariant unit average first applies J^* and then uses inverse phases of the nested paths from a fine site to its block centers. Each path integral is the arithmetic mean of the six coordinate orders. Explicitly, if $c = c_0, c_1, \ldots, c_k = z$ are the nested centers,

$$\varphi_A(z, c) = \exp\big(ie_k \eta \sum_{j<k} \tau A(c_{j+1}, c_j)\big),$$

$$(Q_k(A)\xi)(z) = \eta^3 \sum_{c \in \Delta_z} \varphi_A(z, c)\xi(c),$$

$$(Q_k^T(-A)v)(c) = \varphi_A(z, c)^{-1} v(z).$$

The next average has coefficient L^{-3} and the same six-path phase on the unit lattice. Its algebraic reverse is its transpose at $-A$. Transposes use the indicated bilinear pairings; for real connections they are the corresponding Hermitian adjoints.

Norms and local domains. Distances are periodic sup distances. A block operator uses

$$N_\gamma(T) = \sup_z \sum_{z'} e^{\gamma d(z, z')} \|\mathbf{1}_{\Delta_z} T \mathbf{1}_{\Delta_{z'}}\|_{\infty \to \infty},$$

where the sup norm uses Euclidean spinor norms. Fine kernels use the chamber column measure. For pair outputs the left block restricts the first point and

$$d_X(x, x') = \max(d(c, c'), \eta), \qquad \delta_{\alpha, A} v(x, x') = \frac{e^{ie_k \eta \tau A(c, c')} v(x') - v(x)}{d_X(x, x')^\alpha} \quad (0 < d_X < 1).$$

Transport within a cluster is one. At level zero there are no short pairs; $N_\gamma^{\ker}$ replaces a localized operator norm with the supremum of its normalized kernel on two clusters.

A connection belongs to $\widetilde{\mathcal{G}}_k$ if $A = A_0 + A_1$, A_0 real, $\|A_1\|_\infty < e_k^{-1+\varepsilon}$, and on every $11M$ cube $A_0 = B_0 + d\lambda$ with real B_0, λ and $\|B_0\|_\infty < e_k^{-1+\varepsilon}$. Here $d\lambda(c, c') = \eta^{-1}(\lambda(c') - \lambda(c))$. Write $\|B\|_1 = \|B\|_\infty$, $\|B\|_2 = \|\partial B\|_\infty$, and $\|B\|_3 = \|\delta_\alpha \partial B\|_\infty$, with $\partial_\nu B_\mu(c) = \eta^{-1}[B_\mu(c + \eta e_\nu) - B_\mu(c)]$. The short quotient in the third norm uses $0 < d(c, c') \le 1$. Let R_k have these three norms bounded strictly by $e_k^{-3/4+i\varepsilon}$, $i = 1, 2, 3$. A field in $\widetilde{R}_k$ is $A_0 + A_1$, where $A_1 \in R_k$ and A_0 is real and has these same three bounds after a real gauge transformation on every $(2L + 1)M$ cube. Finally

$$\mathfrak{X}_k = \{C : \|C\|_\infty < e_k^{-1+2\varepsilon}, \quad \|\partial C\|_\infty < e_k^{-1+3\varepsilon}\}.$$

Under $e_k^{1/4} + e_k^{\varepsilon} < 1$, both $\mathfrak{X}_k$ and $\widetilde{R}_k + \mathfrak{X}_k$ are contained in $\widetilde{\mathcal{G}}_k$.

Polymer geometry. An M-polymer is a face-connected union of partition cubes. A net is a connected finite union of periodic images of specified closed straight segments in $\mathbb{R}^3$, allowing a point as a segment. Its length is the sum of the specified segment lengths; $Md_M(X)$ is the infimum over nets and their finite representations that meet every closed cube of X. The used geometric inputs are $|X|_M \le 54(d_M(X)+1)$ and the nonwrapping box criterion: if $2M(d_M(X)+2) < \ell$ and $M(d_M(X)+4) < \ell$, bonds meeting X lift into a product box with side at most $M(d_M(X)+3)$. These facts are imported with their proofs from QEM1 Appendix A; this outline does not assign a second proof of them.

Maxwell input for the reconstructed-field application. To apply the fermion assertions to $H_k^{\mathrm{ph}}A$ and a photon fluctuation Z, import the following actual bounds from QEM1 Section 10: on each enlarged cube $\square^{\dagger}$ of side $(2L+1)M$, a real gauge $\nu_{\square}$ gives $\|H_k^{\mathrm{ph}}A - d\nu_{\square}\|_{i,\square^{\dagger}} \le C_B\|dA\|_{\infty}$ for $i = 1,2,3$; allowed box fluctuations satisfy $\max_i \|Z\|_i \le \frac{1}{2}e_k^{-\varepsilon/2}$; and the margin $A \in \widetilde{R}_k^{-}$ implies $A + tZ \in \widetilde{R}_k$ for $|t| \le \frac{1}{2}e_k^{-3/4+9\varepsilon/2}$. Here $\widetilde{R}_k^{-}$ leaves the margin $e_k^{\varepsilon}e_k^{-3/4+i\varepsilon}$ in its global complex summand. The photon construction and its numerical smallness hypotheses are imported; a fermion conclusion is invoked only after these inclusions have been checked. No photon covariance, positive root, or fluctuation coordinate is constructed here.

Source. QEM1 6790–6886, 7263–7322, 7490–7628, 9072–9107, 10013–10049 and Appendix A. The last paragraph recalls the Maxwell-domain application from 8840–9009, not an extra fermion claim. All line locators refer to the frozen snapshot.

3. S01. The chamber gap and the singular free inverse

Model-specific. Overview Target.

Proof dependencies. F0

Objects and hypotheses. Fix $0 < \rho < 1$. A chamber site is $x = (c, g)$, with $c \in a\mathbb{Z}^3$, $a > 0$, and g a signed permutation matrix. Work on the covering lattice or a torus of side $R \in a\mathbb{N}$, with any unitary character in its three period directions. Pairings use $a^3/48$ per chamber site and the Euclidean norm on $\mathbb{C}^4$; $|\cdot|_2$ on a spin matrix denotes its induced Euclidean operator norm. The internal neighbors are $x^{(i)} = (c, gs_i)$, where s_0 reverses the third coordinate, s_1 exchanges coordinates $2, 3$, and s_2 exchanges $1, 2$. The external neighbor is $x^{(3)} = (c + age_1, gr_{e_1})$, with r_{e_1} the first-coordinate reflection. Put

$$n_0 = e_3, \quad n_1 = (e_2 - e_3)/\sqrt{2}, \quad n_2 = (e_1 - e_2)/\sqrt{2}, \quad n_3 = -e_1, \qquad w = (6, 6\sqrt{2}, 6\sqrt{2}, 6).$$

Use the Hermitian Euclidean representation $\gamma_\mu = \sigma_3 \otimes \sigma_\mu$ and $\gamma_5 = \sigma_1 \otimes 1_2$, with Pauli matrices

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The four-component Dirac matrices satisfy the Clifford relations; $\gamma_5^2 = 1$ and $\gamma_5\gamma_\mu = -\gamma_\mu\gamma_5$. Set $\widehat{d}_i(x) = -gn_i$, $M_i(x) = \frac{1}{2}w_i(\rho - \gamma \cdot \widehat{d}_i(x))$, and

$$D_A f(x) = -a^{-1} \sum_{i=0}^{3} M_i(x)[U_A(x, x^{(i)})f(x^{(i)}) - f(x)].$$

Internal transports are one, and the external transport is $\exp(ie_a aA(c, c+age_1))$. Let $J\xi(c, g) = \xi(c)$, $J^* f(c) = 48^{-1} \sum_g f(c, g)$, $Q = 1 - JJ^*$. Let $\lambda_{\mathrm{int}} > 0$ be the gap of $\sum_{i=0}^{2} w_i(1 - R_i)$ on the mean-zero functions of g; the three generators above make its kernel precisely the constants.

Claim. The bond form $\mathcal{L}_A = \sum_i w_i(1 - U_A R_i)$ satisfies, for real A,

$$(f, \mathcal{L}_A f) \geq \lambda_{\mathrm{int}} \|Qf\|_2^2, \qquad a^{-2}(f, \mathcal{L}_A f) \geq c_* \|\nabla_A J^* f\|_2^2, \quad c_* = \frac{\min(6, \lambda_{\mathrm{int}})}{18}.$$

Here $\nabla_{A,\mu}\xi(c) = a^{-1}[u_A(c, c + ae_\mu)\xi(c + ae_\mu) - \xi(c)]$. For every finite set E of photon sites the same two bounds hold with $\|Qf\|_2$ restricted to sites over E, the positive bond-energy sum restricted to chamber bonds whose two endpoints lie over E, and ∇_A restricted to photon bonds with both endpoints in E. Moreover $QD_AQ + t$ has inverse norm at most $(t + \rho\lambda_{\mathrm{int}}/(2a))^{-1}$ when its denominator is positive. For the zero-field symbol at $a = 1$, split $D_0(p) + m$ by J, Q, write $H_m = D_{hh}^{-1}$, and let $B_m = D_{ll} - D_{lh}H_m D_{hl}$, $0 \leq m \leq 1$. There are positive constants $\theta_0, c_e, \kappa_{\mathrm{strip}}, C_G, c_b$ and finite C_ν, depending only on ρ, such that H_m, B_m are $2\pi\mathbb{Z}^3$-periodic and holomorphic on a neighborhood of $\{|\operatorname{Im} p|_\infty \leq \theta_0\}$. Put $g_h = \rho\lambda_{\mathrm{int}}/2$, $c_{\mathrm{lh}} = \sqrt{3}(1 + \rho)$, $t(p) = \sum_\mu(1 - \cos p_\mu)$, $K_\rho = \rho + 2c_{\mathrm{lh}}^2/g_h$. On real $p \in [-\pi, \pi]^3$,

$$\|H_m(p)\| \leq (g_h + m)^{-1}, \qquad \|D_{lh}(p)\| + \|D_{hl}(p)\| \leq 2c_{\mathrm{lh}}(2t(p))^{1/2},$$

and

$$B_m = m + i\gamma \cdot \sin p + E_m, \quad \|E_m\| \leq K_\rho \sum_\mu (1 - \cos p_\mu),$$

$$\operatorname{Re} B_m \geq m + \rho c_* \sum_\mu (1 - \cos p_\mu),$$

$$|B_m v| \geq c_e(|p|^2 + m^2)^{1/2}|v|, \qquad \|\partial_p^\nu B_m^{-1}\| \leq C_\nu(\rho)(|p|^2 + m^2)^{-(1+|\nu|)/2}.$$

The inverse estimate persists in a strip of width $\kappa_{\mathrm{strip}}m$ when $m > 0$; at $m = 0$ it is asserted away from $p = 0$. The covering Green kernel splits into a label-independent part of size $C_G(1 + |c - c'|_\infty)^{-2}$ and a remainder of size $C_G(1 + |c - c'|_\infty)^{-3}$. A nearest-neighbor difference of the first part has

the latter bound. For $m > 0$, all have the further factor $\exp(-\kappa_{\mathrm{strip}} m|c - c'|_\infty)$. In mesh a, let $G(x, x') = a^{-2} G_{\mu a}(x/a, x'/a)$, $0 \le \mu a \le 1$. Then

$$|G(x, x')|_2 \le 2C_G(a + |c - c'|_\infty)^{-2},$$
$$|G(x^{(i)}, x') - G(x, x')|_2 \le 20 C_G a(a + |c - c'|_\infty)^{-3} \quad (0 \le i \le 3).$$

For distinct $x = (c, g)$, $\widetilde{x} = (\widetilde{c}, \widetilde{g})$, put $d_X(x, \widetilde{x}) = \max(|c - \widetilde{c}|_\infty, a)$. For $0 \le \alpha \le 1$ the full Hölder output is

$$|G(x, x') - G(\widetilde{x}, x')|_2 \le 6240 C_G d_X(x, \widetilde{x})^\alpha [(a + |c - c'|_\infty)^{-2-\alpha} + (a + |\widetilde{c} - c'|_\infty)^{-2-\alpha}].$$

Also prove the free coercivity and cube remainder estimates

$$\|D_0 f\|_2^2 \ge c_b a^{-2}(f, \mathcal{L}_0 f), \qquad \|f - P_E f\|_{2,E}^2 \le C_P \ell_E^2 a^{-2}(f, \mathcal{L}_0 f)_E.$$

Here E is a lattice cube of side $\ell_E \ge a$, $P_E f$ is the constant mean over $E \times O_{\mathrm{cub}}$, and the subscript E uses the local bond sum just defined. One may take $C_P = (8c_*)^{-1} + \lambda_{\mathrm{int}}^{-1}$. All constants are uniform in the mesh, size, character and allowed mass.

Proof work. Use the internal spectral gap and the eight external crossings in each photon direction. Eliminate the heavy component while retaining the real-part form. Combine the Dirac lower bound near zero with the real-part bound away from zero. Differentiate the inverse identity. The general Fourier subproblem is explicit: let T be a $2\pi\mathbb{Z}^3$-periodic finite matrix symbol which is C^N off $p = 0$ on the Brillouin torus and satisfies $\|\partial^\nu T(p)\| \le A|p|_2^{-q-|\nu|}$, $q < 3$, for $|\nu| \le N$, with an integer $N > 3 - q$. Prove Fourier decay $CA(1 + |n|_\infty)^{-(3-q)}$ by a small-ball split and integration by parts. Apply this to the block inverse, then shift contours for positive mass. Obtain the displayed mesh-scaled one-hop and Hölder outputs, including intracluster differences. Finally combine the fiber coercivity with scalar cube Poincaré.

Inputs. Chamber model and spin algebra; finite Fourier analysis. Exports: chamber coercivity, free singular symbol and free-kernel regularity. The Wilson compression D_{ll} is not the Schur complement B_m.

Source. PDF pp. 79–83, §§9.1–9.3, Propositions 9.1–9.4, (9.5)–(9.22); setup repeated here from current TeX lines 6790–6886. Proofs and statements: TeX 6888–7261.

4. S02. Blocked inverses and their local walk expansion

Model-specific. Overview Target.

Proof dependencies. S01

Objects and hypotheses. Let $L \geq 5$ be odd, $\eta = L^{-k}$, and use chamber spinors with measure $\eta^3/48$. The finite side is a positive integer multiple of L, with any unitary period character. The chamber Dirac stencil has four neighbors, three internal and one joining adjacent photon sites. Its inputs are the free coercivity, cube Poincaré and mesh-scaled Green bounds of S01, with constants independent of η. Unit blocks and L-blocks are centered odd cubes. Their covariant averages use the arithmetic mean of the six coordinate-order paths and its reverse phase; denote the induced projections by $\Pi_1(A), \Pi_L(A)$. Put

$$b = 1, \quad b_L = b/L, \quad b_- = b(1 - L^{-1}), \quad b_k = \frac{b(1 - L^{-1})}{1 - L^{-k}} \quad (k \geq 1).$$

For $k \geq 1$, set

$$\begin{aligned} F_y(A) &= D_A + \mu + \alpha_y \Pi_1(A) + \beta_y \Pi_L(A), \\ \alpha_y &= b_k i\gamma_5 y (b_k + i\gamma_5 y)^{-1}, \\ \beta_y &= b_k^2 b_L (b_k + b_L + i\gamma_5 y)^{-1}(b_k + i\gamma_5 y)^{-1}. \end{aligned}$$

Include $y = \infty$ by $\alpha_\infty = b_k, \beta_\infty = 0$. At $k = 0$, instead use $F_y = D_A + \mu + b_L \Pi_L(A) + i\gamma_5 y$, with real y. Here $0 \leq \mu \leq 1$. For unit blocks define the row norm

$$N_\gamma(T) = \sup_z \sum_{z'} e^{\gamma d(z,z')} \|\mathbf{1}_{\Delta_z} T \mathbf{1}_{\Delta_{z'}}\|_{\infty \to \infty}.$$

The sup norm is the supremum of the Euclidean spinor norm. d is the periodic sup distance; kernels use the chamber column measure $\eta^3/48$. For a pair of distinct chamber sites put $d_X(x, x') = \max(d(c, c'), \eta)$ and

$$U_A(x, x') = \exp(ie_k \eta\, \tau A(c, c')), \qquad (\delta_{\alpha,A} v)(x, x') = \frac{U_A(x, x') v(x') - v(x)}{d_X(x, x')^\alpha}, \quad 0 < d_X(x, x') < 1.$$

Here $\tau A(c, c')$ is the mean of the six coordinate-order path sums. If $c = c'$, the transport is one. For pair-valued operators the left block indicator restricts the first point and the output sup runs over all such short pairs. At $k = 0$, N_γ^{ker} is the same weighted row sum with the localized operator norm replaced by the supremum of the normalized matrix kernel over the two clusters. The inverse domain consists of $A = A_0 + A_1$, A_0 real, $\|A_1\|_\infty < e_k^{-1+\varepsilon}$, where on every $11M$ cube $A_0 = B_0 + d\lambda$ with B_0, λ real and $\|B_0\|_\infty < e_k^{-1+\varepsilon}$.

Claim. Fix $0 < \alpha < 1$. There are $C_I, C_5, C_{00}, \gamma_5', a_z > 0$, depending only on L, ρ, α, with $C_5, C_{00} \geq 1$, such that the zero-connection L^2 inverse norm is at most LC_I for positive levels. If f is supported on the unit block $\Delta_{z'}$, $v = F_y(0)^{-1} f$, x lies over Δ_z, and $0 < d_X(x, \widetilde{x}) \leq 1$, then

$$|v(x)|_2 + d_X(x, \widetilde{x})^{-\alpha} |v(x) - v(\widetilde{x})|_2 \leq C_5 \exp(-\gamma_5' d(z, z')) \|f\|_\infty.$$

At zero level,

$$|F_y(0)^{-1}(x, x')|_2 \leq C_{00}(1 + |y|)^{-1} \exp(-a_z d(c, c')).$$

Choose

$$\gamma = \tfrac{1}{2} \min(\gamma_5', a_z) > 0, \qquad a' = \max(C_5, C_{00}) \sum_{n \in \mathbb{Z}^3} \exp(-\gamma |n|_\infty).$$

Use a rate with $\gamma L \leq 1/6$, reducing it and redefining a' by this sum if needed. In particular a' depends only on L, ρ, α, not on M, level, side, character, mass or y. These estimates give $N_\gamma(F_y(0)^{-1}) \leq a' w_k(y)$ and $N_\gamma(\delta_{\alpha,0} F_y(0)^{-1}) \leq a'$; at zero use N_γ^{ker} for the kernel bound. For $e_k \|B\|_\infty \leq \delta$, $3L\delta \leq 1$, $2C_V a' \delta \leq 1$, the perturbed inverse obeys

$$N_\gamma(F_y(B)^{-1}) \leq 2a' w_k(y), \qquad N_\gamma(\delta_{\alpha,B} F_y(B)^{-1}) \leq 18a',$$

where $C_V = 2\exp(1)(42+6b)$, $w_k = 1$ at positive levels and $w_0(y) = (1+|y|)^{-1}$. If $M > 4L$, fold local $3M$ cubes into tori of side $6M$ and patch their inverses with $\sum h_\square^2 = 1$. For $q(p) = 54\pi C_V a' p/M \le 1/2$, length-$n$ walk terms obey

$$\sup_z \sum_{z'} e^{\gamma d(z,z')} \sum_{|\omega|=n} p^{n+1} \|\mathbf{1}_{\Delta_z} S_\omega \mathbf{1}_{\Delta_{z'}}\| \le 16 a' p w_k(y) q(p)^n,$$

$$\sup_z \sum_{z'} e^{\gamma d(z,z')} \sum_{|\omega|=n} p^{n+1} \|\mathbf{1}_{\Delta_z} \delta_{\alpha,A} S_\omega \mathbf{1}_{\Delta_{z'}}\|_{\infty\to\infty} \le 152 a' p q(p)^n.$$

Their sum is the exact inverse, analytic on the stated local-gauge domain, and has row bounds $32a' w_k(y)$ and $304a'$. Require $6Le_0^\varepsilon \le 1$, $4C_V a' e_0^\varepsilon \le 1$ for all $e_k \le e_0$.

Proof work. Prove the averaging-penalty real-part inequalities before invoking the free coercivity. Conjugate by an exponential distance weight. Use the local free Green equation to convert L^2 to sup and Hölder control without mesh losses. Handle the level-zero large-y estimate separately. The complex perturbation is a Neumann series. Verify both local inverse identities on the interior of the folded cube. Bound $[F_y, h_\square]$ by $\pi C_V/M$, then count at most eight initial cutoffs and 27 choices per extra walk step.

Inputs. Chamber coercivity and free-kernel regularity; nested gauge averages. Exports: exact analytic inverse, local inverses, weighted walk sums. The quotient uses $d_X = \max(|c-c'|_\infty, \eta)$, with $0 < d_X < 1$; at $k = 0$ no such pair exists.

Source. PDF pp. 83–89, §§9.4–9.6, Propositions 9.5–9.7, (9.23)–(9.41), especially (9.28). TeX 7263–7808.

5. S05. Weakened fermion walks and component locality

Model-specific. Overview Target.

Objects and hypotheses. Use the chamber field domain, row norm, and M-polymers of F0. The exact local walk terms $S_{y,\omega}$ of S02 have, at length n, the weighted bounds $16a'pw_k(y)q(p)^n$ and $152a'pq(p)^n$ for the short quotient, where $q(p) = 54\pi C_V a'p/M$. For $\omega = (\square_0, \dots, \square_n)$ put

$$X_\omega = \bigcup_{i=0}^{n} \square_i^{\sim 1}, \qquad s_\omega = \prod_{\square \subset X_\omega} s_\square, \qquad |s_\square| \le M^{\alpha_0}, \quad \alpha_0 = 1/54.$$

Here $\square^{\sim 1}$ includes the cube and its one-layer neighborhood. Write S_y for the right walk expansion, S'_y for the left expansion, and $S^{\mathrm{sy}}_y = (S_y + S'_y)/2$ for their mean; $\#$ ranges over these three families. Use the antiperiodic convention of FT for the transpose assertion below. Assume $q(M^{27\alpha_0}) \le 1/2$ and $q(e^{27\kappa'}) \le 1/2$, where $\kappa' > 0$ is fixed.

Claim. The series

$$S^{\#}_y(s, A) = \sum_{\omega} s_\omega S^{\#}_{y,\omega}(A), \qquad S^{\#}_y(X, A) = \sum_{X_\omega = X} S^{\#}_{y,\omega}(A)$$

converge absolutely in the weighted row norms, uniformly in the connection domain and the closed parameter polydisc, and satisfy

$$N_\gamma(S^{\#}_y(s, A)) \le 32a'M^{27\alpha_0} w_k(y),$$
$$N_\gamma(\delta_{\alpha,A} S^{\#}_y(s, A)) \le 304a'M^{27\alpha_0},$$
$$\|\mathbf{1}_{\Delta_z} S^{\#}_y(X, A)\mathbf{1}_{\Delta_{z'}}\| \le 32a'e^{27\kappa'} w_k(y) e^{-\gamma d(z,z') - \kappa' d_M(X)}.$$

At zero level the first and third assertions hold also in the normalized kernel interpretation, with $w_0(y) = (1 + |y|)^{-1}$. At parameter one, $S^{\#}_y(1, A) = S_y(A) = \sum_X S^{\#}_y(X, A)$. The sums are jointly analytic in the connection and interior parameters, continuous on closed parameter polydiscs, and continuous in finite real y.

Each polymer term is supported on X in both arguments and reads only connection bonds with endpoints in X. If the parameters vanish off a union Y of M-cubes, every component of Y is preserved and the operator there depends only on that component's connection and parameters. A nonzero row at (c, g) requires every cube meeting $K_c = \{c' : |c' - c|_\infty \le 2L\}$ to be active; there is an analogous column statement. With $\mathsf{C} = i\gamma_2$,

$$S_y(s, A)^T = \mathsf{C}^{-1} S'_{-y}(s, -A)\mathsf{C}.$$

The symmetrized family consequently has the exact transpose covariance at the neutral values $y = 0, \infty$ for $k \ge 1$ and $y = 0$ for $k = 0$. Gauge covariance holds term by term.

Proof work. A length-n walk has $|X_\omega|_M, d_M(X_\omega) \le 27(n + 1)$. Insert $p = M^{27\alpha_0}$ into the walk estimate and sum its geometric series. For a fixed polymer insert $\mathbf{1}[X_\omega = X] \le e^{-\kappa' d_M(X)}(e^{27\kappa'})^{n+1}$. Retain the support of the first and last cutoffs: the row margin $3M/4 + 2L < 3M/2$ proves the stronger vanishing assertion. A surviving walk lies in one active component, giving local read-dependence. Reversing the factor order proves the transpose identity and explains the need for the symmetrized covariance.

Inputs and use. Only the fine walk estimates and explicit polymer geometry enter this fermion weakening statement. Its later evaluation on photon reconstructions and fluctuations uses the separate Maxwell-domain input recorded in F0. The photon clauses of QEO1 S05 are outside FBO1; this card retains the identifier for its fermion clauses and does not assert those omitted photon estimates.

Source. QEM1 Section 11.1, pp. 102–104, TeX 9091–9248, Proposition 11.1, (11.1)–(11.7). These are the fermion clauses of frozen QEO1 S05.

6. S06. Fine-to-coarse resolvents and local fermion maps

Model-specific. Overview Target.

Proof dependencies. S02, S05

Objects and hypotheses. At mesh $\eta = L^{-k}$, $k \geq 1$, average the 48 chamber labels by J^*, then the nested covariant unit blocks by $Q_k(A)$. Write

$$\mathcal{Q} = Q_k(A)J^*, \quad \mathcal{Q}^\sharp = JQ_k(-A)^T, \quad P_L = Q(-A)^T Q(A), \qquad \mathcal{Q}\mathcal{Q}^\sharp = 1.$$

The maps use normalized scalar phases, so $\mathcal{Q}^\sharp\mathcal{Q} = \Pi_1$, $\mathcal{Q}^\sharp P_L \mathcal{Q} = \Pi_L$; the symbol $\sharp$ is not a Hermitian adjoint for complex A. Take A in the inverse domain: $A = A_0 + A_1$, A_0 real, $\|A_1\|_\infty < e_k^{-1+\varepsilon}$, and on every $11M$ cube $A_0 = B_0 + d\lambda$ for real B_0, λ with $\|B_0\|_\infty < e_k^{-1+\varepsilon}$. The inclusion of the small-field plus external connection domain in this inverse domain is stated explicitly in F0. Use $\mathsf{C} = i\gamma_2$ in the Hermitian Euclidean representation; thus $\bar{H}_k(A) = \mathsf{C}^{-1}H_k(-A)\mathsf{C}$. Let $S_k = (D_A + \mu + b_k\Pi_1)^{-1}$, $0 < \mu \leq 1$, and put

$$H_k = b_k S_k \mathcal{Q}^\sharp, \quad \mathfrak{H}_k = b_k \mathcal{Q} S_k, \quad D_k = b_k - b_k^2 \mathcal{Q} S_k \mathcal{Q}^\sharp, \quad M_k = b_k - D_k.$$

The fine inverse S_y exists with the penalized resolvent parameters of (9.23); set

$$B_{k,y} = (b_k + i\gamma_5 y)^{-1}(1 - P_L) + (b_k + b_L + i\gamma_5 y)^{-1} P_L.$$

Claim. For real y, the coarse operator is invertible and

$$(D_k + b_L P_L + i\gamma_5 y)^{-1} = B_{k,y} + b_k^2 B_{k,y} \mathcal{Q} S_y \mathcal{Q}^\sharp B_{k,y} =: \Gamma_{k,y},$$
$$S_y = S_k + H_k \Gamma_{k,y} \mathfrak{H}_k.$$

This algebra proves coarse invertibility from the fine inverses. With weakened right/left/symmetrized fine families, define

$$H_k(s) = b_k S_\infty(s)\mathcal{Q}^\sharp, \quad \Gamma_k(s) = B_{k,0} + b_k^2 B_{k,0} \mathcal{Q} S_0^{\mathrm{sy}}(s) \mathcal{Q}^\sharp B_{k,0}, \quad M_k(X) = b_k^2 \mathcal{Q} S_\infty^{\mathrm{sy}}(X) \mathcal{Q}^\sharp.$$

For these bounds $a', \gamma > 0$ are S02's defined zero-row constants, with $\gamma L \leq 1/6$; the fine input is exactly the weighted walk and polymer inequalities of S05 at the fixed positive rate κ'. All sup norms use Euclidean spinor norms; normalized chamber kernels use column measure $\eta^3/48$. Specifically H_k and its short covariant quotient satisfy

$$\|H_k(s)\|_{\infty\to\infty} \leq 32\exp(1) b_k a' M^{27\alpha_0}, \qquad \|\delta_{\alpha,A} H_k(s)\|_{\infty\to\infty} \leq 304\exp(1) b_k a' M^{27\alpha_0}.$$

The quotient is $(U_A(x,x')v(x') - v(x))/\max(d(c,c'),\eta)^\alpha$ on $0 < \max(d(c,c'),\eta) < 1$, with the six-path phase and transport one within a cluster. The separate outputs for Γ_k and M_k are

$$|\Gamma_k(s)(z,z')| \leq C_\Gamma M^{27\alpha_0} e^{-\gamma d(z,z')},$$
$$|M_k(X)(z,z')| \leq 32\exp(1) b_k^2 a' e^{27\kappa'} e^{-\gamma d(z,z') - \kappa' d_M(X)},$$
$$C_\Gamma = 2(1 + 2\exp(1))/b_- + 128\exp(1)(1 + 2\exp(1))^2 a'.$$

The assumed fine-family analyticity, component restriction and local read-dependence pass to these families. At $k = 0$ use instead $\Gamma_0 = (D_A + \mu + b_L J P_L J^*)^{-1}$ on all 192 chamber components. Its

weakened normalized kernel has the same decay; $H_0(s)$ is the product of parameters of cubes meeting $K_c = \{|c' - c|_\infty \le 2L\}$, bounded by $M^{8\alpha_0}$. At one it is the identity. There is no M_0.
Proof work. Use both resolvent identities for $Z = b_k^2 \mathcal{Q}^\sharp B_{k,y} \mathcal{Q}$ and multiply the proposed inverse on both sides. Bound the averaging phases by their nested path lengths. Each B has rows and columns in one L-block; compose their row sums and retain the distance shift $2L$. Nested blocks and the fine component restriction give locality. Transposition exchanges right and left walk expansions, explaining the symmetrized covariance.
Inputs. Exact and weakened fine inverse estimates, including their support and component statements; normalized gauge averages. Exports: coarse resolvent, field maps, local Dirac kernels. $\mathfrak{H}_k$ is the right resolvent map; the barred field map is separately $\bar{H}_k(A) = \mathsf{C}^{-1} H_k(-A)\mathsf{C}$.
Source. PDF pp. 104–107, §§11.2–11.3, Propositions 11.2–11.3, (11.8)–(11.19). TeX 9249–9513.

7. S07. Weighted Berezin integration and exact block completion

Interface. Overview Target.

Objects and hypotheses. Let I be a finite ordered mode set with positive weights $\nu(t)$, and independent generators $W(t), \bar{W}(t)$. An operator kernel acts using the column weight $\nu(t')$; its quadratic form is $\sum_{t,t'} \nu(t)\nu(t')\bar{W}(t)O(t,t')W(t')$. Define $\mathrm{D}W$ as extraction of the coefficient of $\prod_t W(t)\bar{W}(t)$, multiplied by $\prod_t \nu(t)^{-1}$. For invertible C, put

$$\mathrm{d}\mu_C(W) = (\det C)\,\mathrm{D}W \exp[-\langle \bar{W}, C^{-1}W\rangle_\nu].$$

For any finite families of Grassmann generators give an ordered monomial the product of its generator weights, and a polynomial the sum of the moduli of its coefficients times those weights. This is equivalently the absolute antisymmetric-kernel norm with the index measures and the factors $1/n!$.
Claim. The norm is submultiplicative. With $w_W = \max_t \nu(t)^{-1/2}$,

$$\left\| \int \mathrm{d}\mu_I(W) F \right\|_w \le \|F\|_{w,w_W},$$

$$\int \mathrm{d}\mu_C(W) F(\bar{W}, W) = \int \mathrm{d}\mu_I(W) F(\bar{W}, CW),$$

$$\int \mathrm{d}\mu_C(W) W(t)\bar{W}(t') = C(t,t'), \qquad \int \mathrm{D}W\, e^{-\langle \bar{W}, OW\rangle_\nu} = \det O.$$

The last formula does not require O invertible. Disjoint mode sets factor, with external generators retained as coefficients. For linear odd fields $v_1, \dots, v_{2r}$, prove the Pfaffian Wick formula; odd moments vanish. If a linear substitution sends family φ_i to $\sum_j O_{ij}\theta_j$, its norm is at most one whenever

$$\sum_j \left(\sup_t \sum_s |O_{ij}(t,s)| \nu'_j(s) \right) w'_j \le w_i$$

for each occurring input row.

For two finite spinor spaces with their specified pairings, let $K = D + b_L A^\sharp A$ be invertible, $\Gamma = K^{-1}$, and set

$$\Psi^c = b_L \Gamma A^\sharp \Phi, \quad \bar{\Psi}^c = b_L \Gamma^T A^T \bar{\Phi}, \quad D' = b_L - b_L^2 A \Gamma A^\sharp.$$

Put $s = b_L A^\sharp \Phi$, $\bar{s} = b_L A^T \bar{\Phi}$. Then completion of the block quadratic form gives

$$\int \mathrm{D}\Psi \, \exp\{-\langle \bar{\Psi}, K\Psi\rangle + \langle \bar{\Psi}, s\rangle + \langle \bar{s}, \Psi\rangle - b_L \langle \bar{\Phi}, \Phi\rangle\} F(\bar{\Psi}, \Psi)$$

$$= \det K\, e^{-\langle \bar{\Phi}, D'\Phi\rangle} \int \mathrm{d}\mu_I(W) F(\bar{\Psi}^c + \bar{W}, \Psi^c + \Gamma W).$$

Here $A^\sharp$ is the specified algebraic reverse map. In the chamber application it is the transpose of the averaging map at the reversed connection. No Hermitian conjugation of Grassmann fields is used.
Proof work. Compute each paired monomial in the identity Gaussian. Its modulus is exactly covered by w_W. A linear change of the unbarred variables multiplies the top coefficient by the operator determinant. Left Grassmann differentiation and integration by parts fix the Wick signs. Bound images of generators for the substitution estimate. Translate by the displayed means, keeping odd factors in their original order, and cancel both cross terms.
Inputs. Finite exterior algebra and determinants only. Exports: weighted moments, substitutions, exact Gaussian block identity. In the application $\nu = 1/48$ at level zero, so $w_W = \sqrt{48}$ and the identity two-point kernel is $48\delta_{tt'}$. At positive levels $\nu = 1$.
Source. PDF pp. 107–110, §§11.4–11.5, Proposition 11.4, (11.20)–(11.29). TeX 9514–9754.

8. S08. A local determinant logarithm and its small-field gain

Model-specific. Overview Target.

Proof dependencies. F0, S05, S06, S07

Objects and hypotheses. Use fermions antiperiodic in each coordinate. On a torus partitioned into M-cubes, a polymer X is face-connected and $Md_M(X)$ is its minimal represented-net length. Assume $|X|_M \le 54(d_M(X)+1)$ and the nonwrapping lift/box criteria. Put $\eta = L^{-k}$, $0<\alpha<1$, $b=1$, $b_L = b/L$, $b_k = b(1-L^{-1})/(1-L^{-k})$ for $k \ge 1$. Let $C_w, \gamma_w, \kappa, \kappa_0$ be fixed positive constants. The fine-block norm below uses the supremum of the Euclidean spinor norm. Determinant traces use scalar component row sums. Choose C_w to bound both conventions, enlarging an Euclidean bound by at most two for the component sup norm. Normalized fine chamber kernels use column measure $\eta^3/48$. At positive levels the unit-spinor operators satisfy

$$\delta Z_k(A) = \det(D_k + b_L P_L), \quad B_{k,y} = (b_k + i\gamma_5 y)^{-1}(1-P_L) + (b_k + b_L + i\gamma_5 y)^{-1} P_L.$$

Here P_L is a covariant block projection, $\mathcal{Q}\mathcal{Q}^\sharp = 1$, and the fine polymer inverses obey

$$\|\mathbf{1}_{\Delta_z} S_{k,y}(X,A)\mathbf{1}_{\Delta_{z'}}\| \le C_w e^{-\gamma_w d(z,z') - \kappa_w d_M(X)}, \qquad \kappa_w = \kappa + \kappa_0 + 3.$$

At zero use $\delta Z_0 = \det F_0$ on all chamber modes, with measure $1/48$, and the normalized kernel bound has the further factor $(1+y)^{-1}$. Assume analyticity, gauge and step covariance, the transpose/reality identities, $0<\mu\le 1$, and the proved inverse domain consisting of locally real-gauge-small fields.

Claim. Define

$$\widetilde{E}_k(X,A) = -ib_k^2 \int_0^\infty \mathrm{Tr}[\gamma_5 B_{k,y} \mathcal{Q} S_{k,y}(X,A) \mathcal{Q}^\sharp B_{k,y}]\,\mathrm{d}y \qquad (k \ge 1),$$

$$\widetilde{E}_0(X,A) = \int_1^\infty \mathrm{Tr}[F_0 S_{0,y}(X,A)]\,\frac{\mathrm{d}y}{y} - i\int_0^1 \mathrm{Tr}[\gamma_5 S_{0,y}(X,A)]\,\mathrm{d}y.$$

The chamber trace includes its $1/48$ measure. Let E_k^d be the average of $\widetilde{E}_k(A)$, $\widetilde{E}_k(-A)$, $\overline{\widetilde{E}_k(\bar{A})}$, $\overline{\widetilde{E}_k(-\bar{A})}$; set $E_k^{\det}(X,A) = E_k^d(X,A) - E_k^d(X,0)$. These are absolutely convergent, local analytic functions and

$$|\widetilde{E}_k(X,A)|,\ |E_k^d(X,A)| \le C_d M^3 e^{-(\kappa+1)d_M(X)},$$

$$\delta Z_k(A) = \exp\{c_k^d \ell^3 + \sum_X \widetilde{E}_k(X,A)\}, \qquad \frac{\delta Z_k(A)}{\delta Z_k(0)} = \exp\{\sum_X E_k^{\det}(X,A)\}.$$

Here $c_0^d = 0$, $c_k^d = 4[(1-L^{-3})\log b_k + L^{-3}\log(b_k+b_L)]$, and

$$C_d = 16(54)C_w \max\{\tfrac{\pi}{2} b \exp(1)(1+2\exp(1)), 2 + 58\exp(1) + b\exp(1)\}.$$

Each E_k^d reads only bonds meeting X, is real-gauge invariant and even, and satisfies the analytic reality identity.

For the small-field assertion define the norms explicitly by

$$\|B\|_1 = \sup_{c,\mu} |B_\mu(c)|, \qquad \|B\|_2 = \sup_{c,\mu,\nu} |\partial_\nu B_\mu(c)|,$$

$$\|B\|_3 = \sup_{\mu,\nu} \sup_{0<d(c,c')\le 1} \frac{|\partial_\nu B_\mu(c') - \partial_\nu B_\mu(c)|}{d(c,c')^\alpha},$$

$$\partial_\nu B_\mu(c) = \eta^{-1}[B_\mu(c+\eta e_\nu) - B_\mu(c)].$$

The distance is periodic sup distance; local norms use only bonds and difference cells inside their stated set. For A with local-real-gauge plus global complex bounds $(e_k^{-3/4+\varepsilon}, e_k^{-3/4+2\varepsilon}, e_k^{-3/4+3\varepsilon})$ on its three norms, and $\|C\|_\infty < e_k^{-1+2\varepsilon}$, $\|\partial C\|_\infty < e_k^{-1+3\varepsilon}$, prove

$$|E_k^d(X, A+C) - E_k^d(X,C)| \le 8 C_d M^3 e_k^{1/4} e^{-\kappa d_M(X)}.$$

Require $v = -\log e_k \geq 9$, $e_k^{\varepsilon} \leq 1/4$, $8M(v/4+4)e_k^{\varepsilon} \leq 1$, $\ell \geq Mv^2$, and $e_k^{1/4} + e_k^{\varepsilon} < 1$.
Proof work. At real A, use the real eigenvalues of the Hermitian matrix $K\gamma_5$ to integrate scalar logarithms; retain the ranks of $P_L, 1-P_L$ and the integral phases. Two B factors give $(b_k^2+y^2)^{-1}$. At zero keep the split integration intervals and the $(1+y)^{-1}$ factor. Analytic continuation and continuous $2\pi i\mathbb{Z}$-valued differences justify the symmetrized quotient without a global logarithm branch. For the gain, large polymers pay $e_k^{1/4}$ from their decay. Lift the remaining polymers into a nonwrapping product box, subtract a real path-sum gauge, extend the sup-small local field by zero and apply Cauchy's difference formula on a disk of radius $\frac{1}{3}e_k^{-1/4}$, with the external slot fixed.
Inputs. Local fine/coarse resolvent kernels; weighted trace conventions; polymer lifts and geometry.
Exports: local determinant primitives and their $e_k^{1/4}$ difference gain.
Source. PDF pp. 110–113, §§11.6–11.8, Propositions 11.5–11.6, (11.30)–(11.38). TeX 9755–10116.